

NaturaNest

Pure. Natural. Healthy.

Project Phase 2 — Technical Report

CSCI390: Web Programming | Spring 2025–2026

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Submitted

May 26, 2026

Due: June 1, 2026

Live Demo

[Mahdi-Yassine.github.io
/natura-nest-app](https://Mahdi-Yassine.github.io/natura-nest-app)

Abstract

NaturaNest is a fully responsive, multi-page frontend web application built with ReactJS and Bootstrap 5. It serves as a digital storefront for a natural and healthy food brand, offering an immersive browsing experience across four main pages: Home, Products, About Us, and Contact.

Developed as Phase 2 of the CSCI390 Web Programming course, this phase introduces a fully functional React-based single-page application (SPA) using React Router for client-side navigation, component-based architecture for code reusability, and React Hooks for state management — powering features such as product filtering and contact form handling.

The application is version-controlled with Git and deployed live on GitHub Pages. The codebase follows modern React conventions: functional components, `useState`, and modular scoped CSS with BEM naming.

System Design

1. Application Architecture

NaturaNest follows a Component-Based Architecture (CBA) — the standard paradigm for ReactJS applications. The UI is decomposed into small, reusable, self-contained components across three layers:

- Entry Layer — `index.js` bootstraps React into the DOM and imports Bootstrap styles.
- Routing Layer — `App.js` configures `HashRouter` and maps paths to page components.
- Presentation Layer — Page components assemble reusable UI components per screen.

2. Folder Structure

The project follows a clean, feature-based directory layout:

```
natura-nest-app/  
  src/  
    assets/      # Product & team images (JPG)  
    components/  # Reusable UI components  
      FeatureBox/ # 'Why Choose Us?' cards  
      FilterBar/  # Category filter buttons  
      Footer/     # Site-wide footer  
      Hero/       # Landing hero banner  
      Navbar/     # Sticky responsive navigation  
      PageBanner/ # Page title banner  
      ProductCard/ # Individual product card  
      ProductList/ # Responsive grid of ProductCards  
      TeamCard/   # Team member profile card  
    data/  
      products.js # Static catalogue – 8 products  
    pages/  
      About/      # Story, Mission/Vision/Values, Team  
      Contact/    # Info panel + submission form  
      Home/       # Hero + Features + Best Sellers  
      Products/   # Filterable full catalogue  
    App.js        # Root component + router config  
    index.js      # ReactDOM entry point
```

3. Routing

Client-side routing uses `react-router-dom v7` with `HashRouter` (# URLs), required for correct behaviour on GitHub Pages.

Route	Component	Description
/	Home	Landing — Hero, Features, Best Sellers

/products	Products	Full catalogue with category filtering
/about	About	Brand story, Mission/Vision/Values, Team
/contact	Contact	Contact info + message form

4. Component Breakdown

Component	Responsibility
Navbar	Sticky top nav. NavLink active-link highlighting. Hamburger toggle via useState.
Hero	Full-width banner with headline, tagline, and 'Shop Now' CTA to /products.
FeatureBox	Three brand value cards: 100% Natural, Locally Made, Healthy Living.
FilterBar	Category filter buttons. Receives activeFilter and onFilterChange as props.
ProductCard	Shows product image, name, description, price. Conditional Add to Cart button.
ProductList	Maps products array to a responsive Bootstrap grid of ProductCards.
PageBanner	Reusable page header (title + subtitle). Used on Products, About, Contact.
TeamCard	Circular team photo, name, and role — rendered on the About page.
Footer	Copyright footer rendered on every page.

5. State Management

The app uses React's built-in useState hook — no external state library needed:

- Products page: activeFilter drives real-time category filtering via Array.filter().
- Contact page: formData tracks all four fields; successMsg shows confirmation on submit.
- Navbar: menuOpen toggles the mobile hamburger menu open/closed.

6. Responsive Design

Responsiveness is achieved through Bootstrap 5's grid (col-12, col-sm-6, col-md-4, col-lg-3) combined with scoped CSS. The Navbar collapses to a hamburger on mobile. All product grids, team cards, and info panels reflow from single-column on mobile to multi-column on desktop.

Technologies Used

Technology	Version	Purpose
React	19.2.6	Core UI library — components, Hooks, SPA lifecycle
React DOM	19.2.6	Mounts the React tree into the HTML document
React Router DOM	7.15.1	Client-side routing: HashRouter, Routes, NavLink
Bootstrap	5.3.8	Responsive grid, utility classes, form controls
React Scripts (CRA)	5.0.1	Dev server, build pipeline, project bootstrapping
gh-pages	6.3.0	One-command GitHub Pages deployment (npm run deploy)
Web Vitals	2.1.4	Core performance monitoring
Git / GitHub	—	Version control and source code hosting
GitHub Pages	—	Free static site deployment host
CSS (scoped)	—	Component-level styling with BEM naming convention

Key Code Snippets

1. App.js — Root Router Configuration

HashRouter wraps four routes; Navbar and Footer persist on every page:

```
function App() {
  return (
    <Router>
      <div className='app'>
        <Navbar />
        <main className='main-content'>
          <Routes>
            <Route path='/'      element={<Home />}    />
            <Route path='/products' element={<Products />} />
            <Route path='/about'  element={<About />}   />
            <Route path='/contact' element={<Contact />}  />
          </Routes>
        </main>
        <Footer />
      </div>
    </Router>
  );
}
```

2. Products.js — Category Filtering

activeFilter state drives Array.filter() for real-time product filtering:

```
function Products() {
  const [activeFilter, setActiveFilter] = useState('all');

  const filtered =
    activeFilter === 'all'
      ? products
      : products.filter(p => p.category === activeFilter);

  return (
    <div className='nn-products-page'>
      <FilterBar activeFilter={activeFilter}
        onFilterChange={setActiveFilter} />
      <ProductList products={filtered} showButton
        onAddToCart={handleAddToCart} />
    </div>
  );
}
```

3. Navbar.js — Mobile Hamburger Toggle

useState manages collapse state; NavLink isActive drives the active style class:

```
function Navbar() {
  const [menuOpen, setMenuOpen] = useState(false);
  const close = () => setMenuOpen(false);
  return (
    <nav className='nn-navbar sticky-top'>
      <NavLink to='/' onClick={close}>NaturaNest</NavLink>
      <button onClick={() => setMenuOpen(prev => !prev)}>
        <span className='navbar-toggler-icon' />
      </button>
      <div className={`collapse${menuOpen ? ' show' : ''}`}>
        {navLinks.map(({ to, label }) => (
          <NavLink to={to}
            className={({ isActive }) =>
              `nav-link${isActive ? ' active' : ''}`
            onClick={close}>{label}</NavLink>
        ))}
      </div>
    </nav>
  );
}
```

4. Contact.js — Controlled Form with Success Feedback

All inputs bound to formData state; submit resets form and shows personalised confirmation:

```
const initialForm = { name:'', email:'', subject:'', message:'' };

function Contact() {
  const [formData, setFormData] = useState(initialForm);
  const [successMsg, setSuccessMsg] = useState('');

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData(prev => ({ ...prev, [name]: value }));
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    setSuccessMsg(`Thank you, ${formData.name}! Message sent.`);
    setFormData(initialForm);
  };
}
```

5. products.js — Static Data Layer

8 products in one exported array — each with id, name, description, price, image, category:

```
const products = [
  { id:1, name:'Natural Peanut Butter', price:'8.99',
```

```
    image: natural, category:'butter',
    description:'Creamy, 100% roasted peanuts, no additives.' },
{ id:3, name:'Raw Honey', price:'12.50',
  image: honey, category:'sweet' },
{ id:5, name:'Organic Almonds', price:'10.00',
  image: almondNat, category:'snack' },
// ... 5 more across butter / sweet / snack
];
export default products;
```

UI Screenshots

The following screenshots show all four pages of the live NaturaNest application as deployed on GitHub Pages, demonstrating the consistent forest-green design, responsive layout, and interactive features.

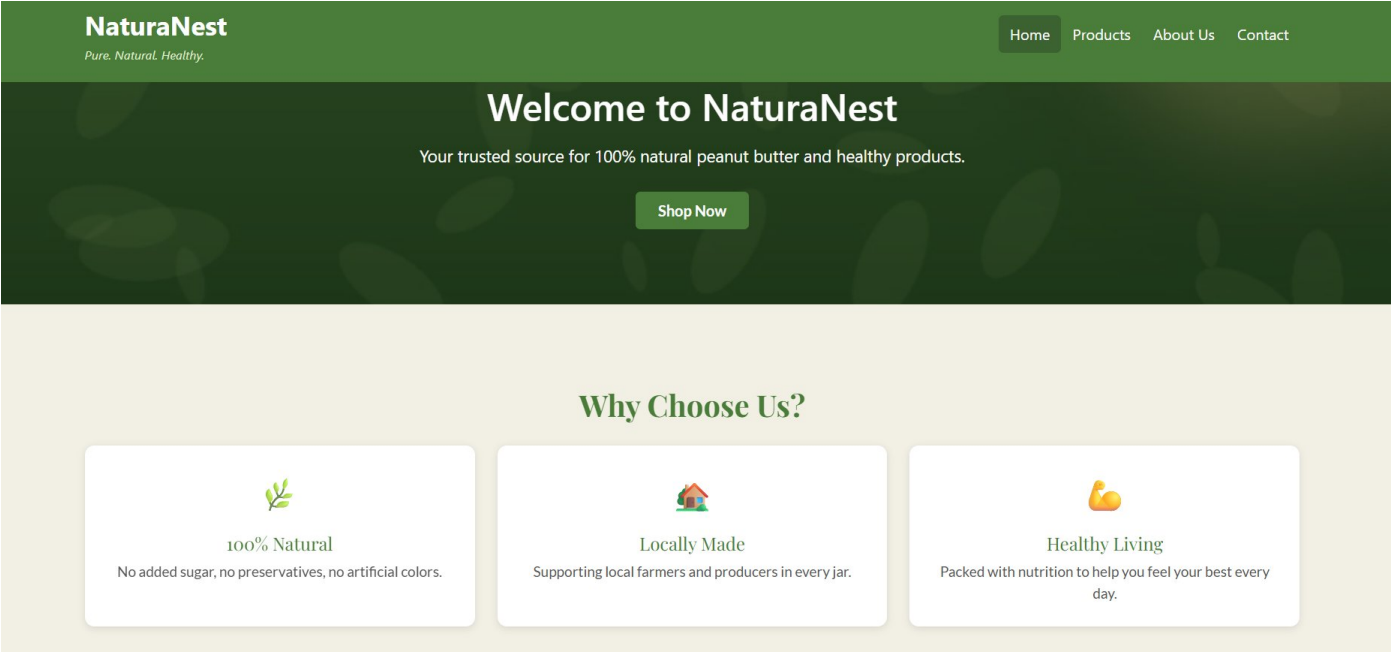


Figure 1 — Home Page: Hero section and 'Why Choose Us?' feature cards

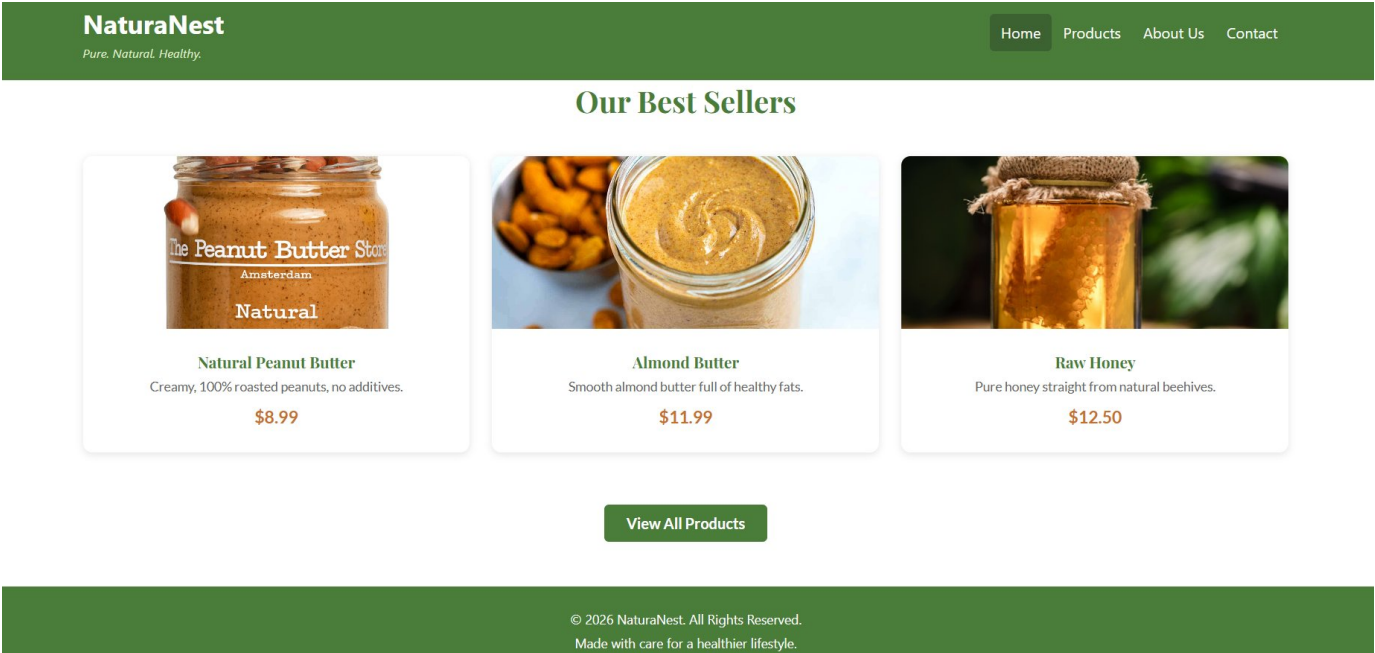


Figure 2 — Home Page: Best Sellers product grid and footer

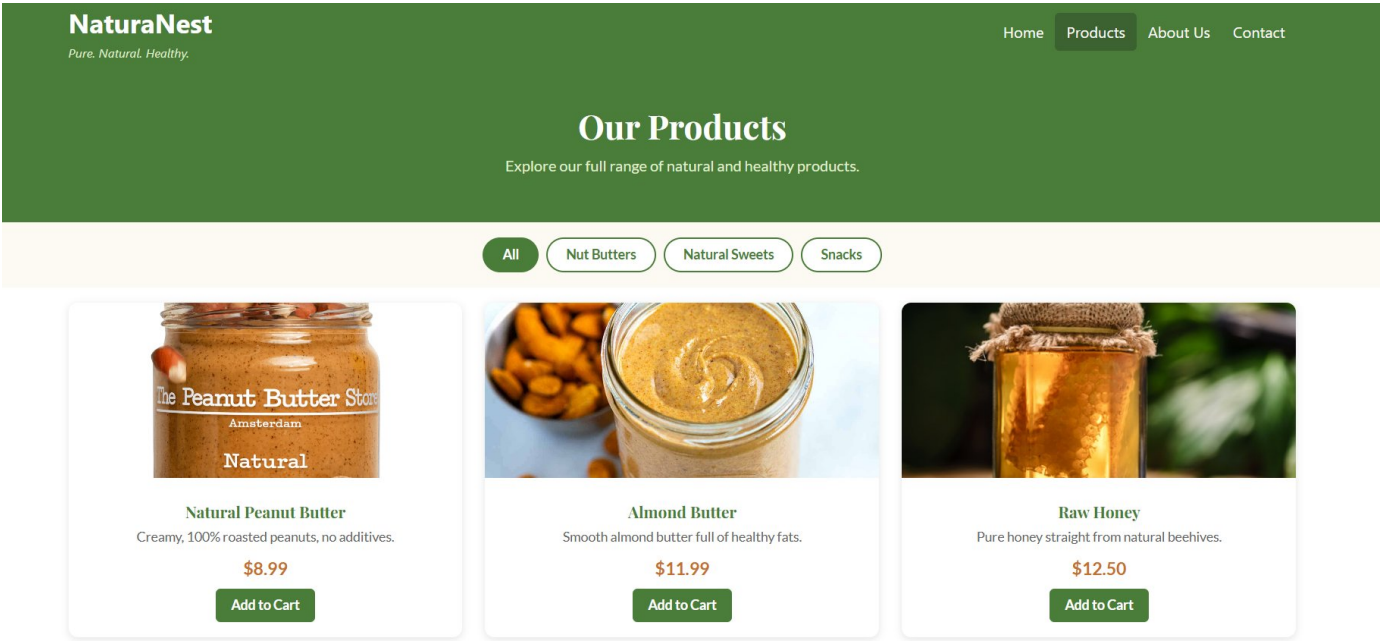


Figure 3 — Products Page: Category filter bar (All) and full product grid

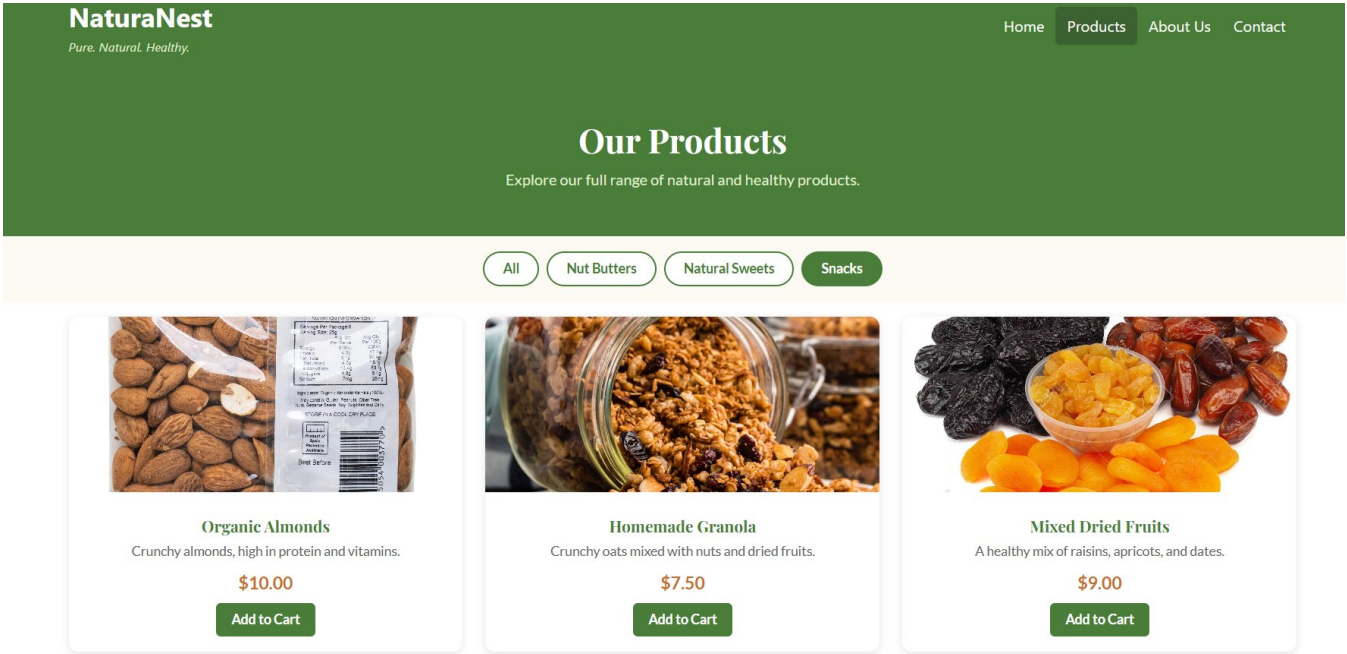


Figure 4 — Products Page: Snacks category filter active

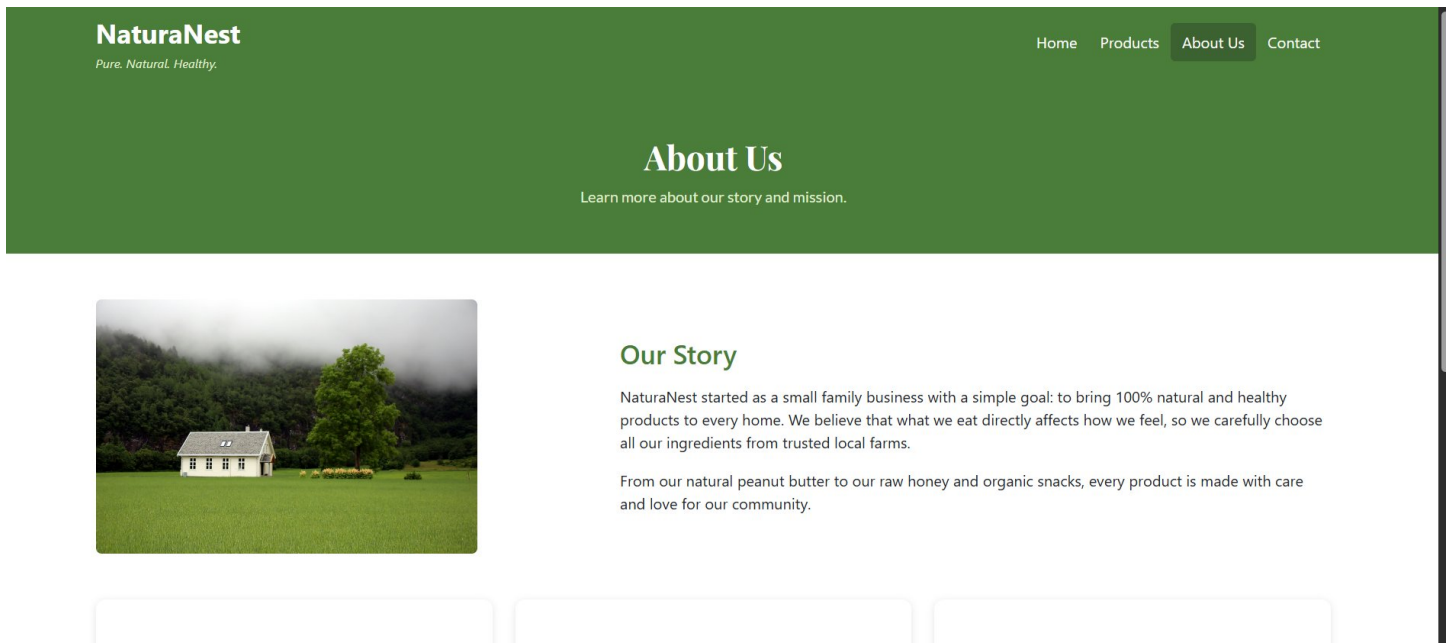


Figure 5 — About Us Page: Brand story section with image

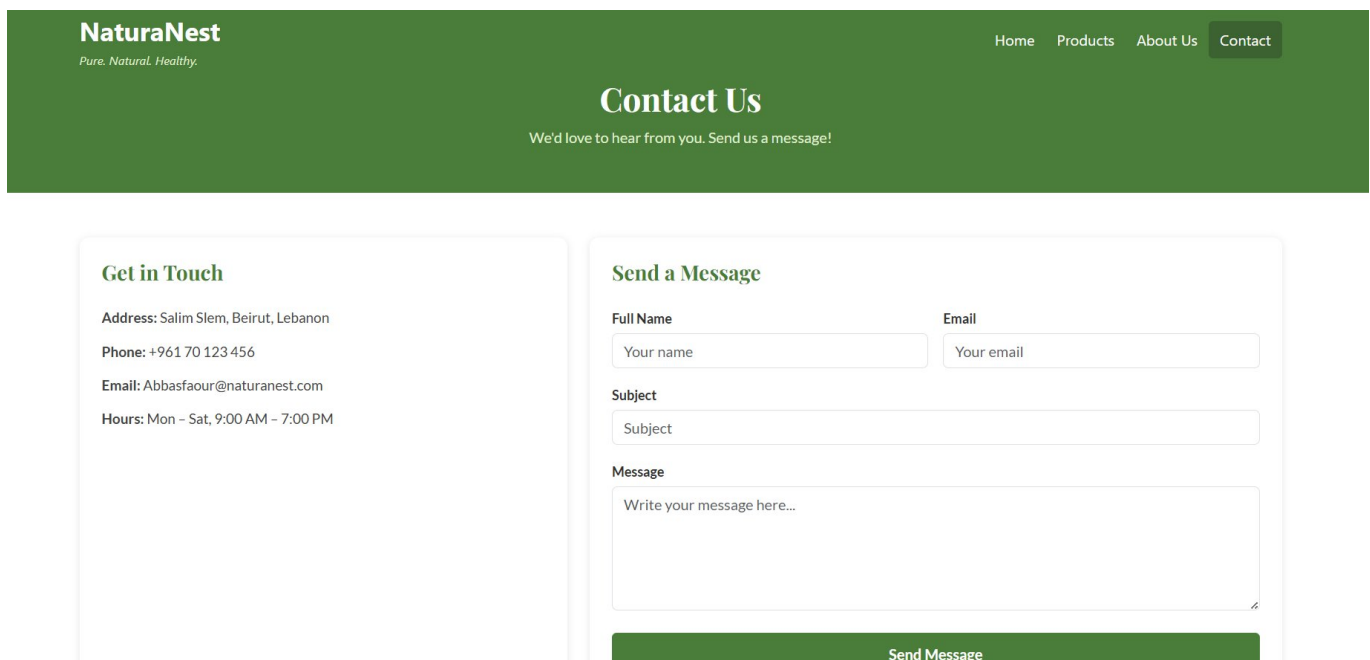


Figure 6 — Contact Page: Info panel and message form

Features Overview

Feature	Details
Multi-page SPA	4 pages with zero full reloads via React Router HashRouter
Responsive Layout	Mobile-first Bootstrap 5 grid — works on all screen sizes
Product Catalogue	8 natural food products displayed in a responsive card grid
Category Filtering	Real-time filter: All / Nut Butters / Natural Sweets / Snacks
Add to Cart (UI)	Per-product button with alert feedback on the Products page
Best Sellers	Home page auto-shows first 3 products as featured best sellers
Brand Story & Team	About page: story, mission/vision/values cards, team profiles
Contact Form	Fully controlled 4-field form with personalised success message
Sticky Navigation	Navbar stays visible on scroll; active link highlighted
Hamburger Menu	Collapsible mobile menu managed with useState hook
GitHub Pages Deploy	Live public URL via gh-pages npm script (npm run deploy)
Version Control	Full commit history tracked on GitHub master branch

Setup & Running Instructions

Prerequisites

- Node.js v18 or later
- npm v9 or later
- Git

Local Development

```
# 1. Clone the repository
git clone https://github.com/Mahdi-Yassine/natura-nest-app.git
cd natura-nest-app

# 2. Install dependencies
npm install

# 3. Start the development server
npm start

# App opens at http://localhost:3000
```

Production Build & Deployment

```
# Build optimised static files
npm run build

# Deploy to GitHub Pages
npm run deploy

# Live URL:
# https://Mahdi-Yassine.github.io/natura-nest-app
```

Group Contribution Statement

As required by submission guidelines, the table below details each member's contributions:

Team Member	Contributions
Mahdi Yassine	Project setup (CRA), App.js router config, Navbar, Hero, Home page, ProductCard & ProductList components, products data file, GitHub Pages deployment, Git repository management.
Mahdi Aljabali	Products page with FilterBar and filtering logic, Contact page with controlled form and success feedback, About page (story, mission/vision/values, team), FeatureBox, PageBanner, TeamCard, Footer components, responsive CSS styling across all components.

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